PROJECT SEVA SETU: FIXING THE LEGACY

Briefing Synthesis, Issue Diagnosis & Strategic Remediation Roadmap

Build for Bharat Fellowship — Citizen-Facing Pension IT Modernization

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Context & Role: Appointed as a *Build for Bharat Fellow* in a state government IT administration, you inherit **Seva Setu**—a production pension portal serving over 100,000 senior citizens with 6 months of active server logs. The vendor contract terminated with zero documentation. Based on the Deputy Commissioner's briefing and portal mandate, this document details all discussed operational/architectural defects and outlines the strategic solutions expected from you.

1. Exhaustive Breakdown of Problems Discussed by the Deputy Commissioner

Area / Issue	Observed Symptom & Field Impact	Probable Technical Root Cause
1. UTF-8 / Indic Script Encoding Failure	An elderly applicant walked 2 hours to report his Assamese name appearing as **'????' (4 question marks)** on acknowledgement slips. Others experience names cut in half.	• Database charset/collation configured as latin1/ascii instead of utf8mb4. • Truncation indicates fixed-length VARCHAR columns storing multi-byte UTF-8 graphemes exceeding byte boundaries.
2. Absence of Application Editing / Cancellation Flow	Helpdesk is overwhelmed with citizens begging to correct minor typos. No 'edit', 'draft', or 'cancel' option exists, forcing manual record deletions and complete resubmissions.	• Strict, one-way transactional write pipeline with no 'Draft' / 'Under Review' mutable states. • Schema enforces unique identity constraints preventing re-application without hard administrative deletions.
3. Broken RTPS Deemed Approval Pipeline	Under the *Right to Public Services Act (RTPS)*, applications pending >15 days must automatically flip to **'Deemed Approved'**. This automated overnight job abruptly stopped functioning around Holi.	• Scheduled Cron job / daemon worker crashed or encountered unhandled exceptions. • Timezone misconfiguration (UTC vs IST), broken SQL status migration script, or lock contention during batch updates.
4. Disbursement Cyclical Denial of Service (DoS)	2–3 days prior to the monthly pension disbursement cutoff, the portal grinds to a complete crawl. Staff shrug off severe latency as an unfixable seasonal occurrence.	• Unindexed database queries executed on high-frequency disbursement verification runs. • N+1 query loops, missing Redis caching layers, connection pool exhaustion, or CPU-bound PDF generation locks.
5. Critical PII Leakage & Privacy Breach	Immediately after applying, elderly citizens receive unsolicited sales calls and home sales visits, indicating active live data leakage.	• Unauthenticated public API endpoints exposing pensioner PII (IDOR / Broken Object-Level Auth). • Insecure direct database exposure, public S3 buckets, compromised vendor credentials, or third-party tracking scripts.
6. Hostile UX / Exclusionary Digital Form	Eligible seniors fail to navigate the digital application. Exploitative middlemen in local bazaars charge high fees to complete forms that citizens cannot decipher.	• Overly rigid regex validation, non-accessible form architecture, absence of vernacular language support, lack of mobile-responsive touch targets, and zero assistive error prompts.

2. Governing Statutory & Business Rules Stated in Briefing

• **Target Beneficiary Demographic:** State residents aged **60 years and above** strictly evaluated on the date of application.
• **Application Cardinality:** Exactly **one application per person** (enforced via unique demographic/Aadhaar/eKYC constraints).
• **Statutory RTPS Timeline:** Mandatory decision window of **15 calendar days**. Applications untouched past this threshold become deemed approved by legal right.
• **Citizen UI Truth:** Unprocessed files exceeding the 15-day statutory window must explicitly reflect an **'Approved'** status on citizen acknowledgement screens.
• **System Ingestion Cutoff:** Daily portal submission intake closes promptly at **midnight (12:00 AM IST)**.

3. Solutions & Deliverables Expected from the Fellow

Expected Solution / Deliverable	Technical Implementation Details & Fellowship Mandate
1. Production Server Hosting & Deployment	• Stand up the legacy codebase, 100k-record database, and server environment on an independently hosted public server. • Maintain a verified **Deployed Public URL** reachable continuously for up to ~30 days until evaluation finishes.
2. Database & Internationalization Repair	• Migrate SQL tables and character collations to utf8mb4_unicode_ci across storage, schema, and API serialization. • Adjust VARCHAR column thresholds to accommodate Indic (Assamese/Bengali/Devanagari) multi-byte graphemes.
3. RTPS Automation Engine Restoration	• Debug and reactivate the automated overnight batch processor (Cron/Celery/systemd service) that failed around March. • Correct timezone offsets to ensure reliable IST midnight evaluations and guarantee status reflection on citizen portals.
4. API Security Hardening & Zero-Leakage Audit	• Restrict endpoint visibility: audit API access with JWT/session guards to prevent unauthenticated data harvesting. • Sanitize application logging to eliminate cleartext PII in log files, stop external leak vectors, and neutralize predatory outreach.
5. Performance Tuning for Disbursement Peaks	• Analyze slow query execution plans (EXPLAIN ANALYZE); establish compound indices on status, date, and citizen ID. • Introduce Redis caching or read-replicas for disbursement verification queries to prevent system-wide lockouts.
6. Citizen-Centric UX & Error Recovery Flow	• Introduce editable application drafts, pre-submission confirmation modals, and self-serve correction/cancellation windows. • Simplify form layout for digital literacy, eliminating dependence on predatory village intermediaries.
7. Video Debrief Submission (Evaluation Criteria)	• **No formal written PDF report expected by the DC**—record a short screen & face video. • Articulate the **Top 5 high-impact fixes** in ranked order with justification, plus explicit rationale for bugs deliberately left alone.

4. Recommended Priority Matrix for the Top 5 Video Fixes

In your recorded video debrief, frame your top 5 technical interventions based on *civic harm reduction* and *legal compliance*:
1. Fix PII Leakage / API Vulnerability: Highest ethical and privacy priority to immediately halt harassment of vulnerable seniors.
2. Restore RTPS Deemed Approval Engine: Essential statutory obligation preventing administrative penalties and legal appeals.
3. Fix Assamese UTF-8 Character Truncation: Core dignity and civic access fix ensuring valid legal acknowledgement slips.
4. Optimize Pre-Disbursement Server Bottlenecks: Infrastructure stabilization to prevent portal collapse during critical payout periods.
5. Implement Self-Serve Correction / Cancellation Window: Relieves catastrophic helpdesk volume and reduces citizen anxiety.
What to deliberately leave alone: Major architectural overhauls, deep UI cosmetic redesigns, or vendor-originated multi-tenancy rewrite—focus strictly on stabilizing high-impact citizen workflows.